

RESEARCH ARTICLE

Spectrum Anomaly Detection Using Deep Neural Networks: A Wireless Signal Perspective

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This work was supported in part by the Information and Communication Technology (ICT) Research and Development Program of Ministry of Science and ICT (MSIT)/Institute for Information communication Technology Planning and Evaluation (IITP), Development of the Next-Generation Spectrum Monitoring Platform Based on Big-Data and Artificial Intelligence (AI) Technology under Grant RS-2023-00229541; and in part by the Regional Innovation Strategy (RIS) through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (MOE) under Grant 2021RIS-004.

ABSTRACT In modern society, LTE and 5G communications serve as essential infrastructures, making the stability of communication networks a critical challenge. However, advancements in wireless communication devices and technologies can lead to the transmission of illegal signals, device malfunctions, and radio interference, potentially causing degradation of communication performance. This study proposes a wideband spectrum monitoring method for LTE and 5G communication bands used in cellular networks by utilizing object detection techniques, thereby ensuring stable network performance and security. The proposed network accounts for the temporal and frequency characteristics present in signals on the spectrogram and is designed to exhibit high detection performance even for low-power signals. It achieved an mean Average Precision(mAP)50 performance of 80% at an Interference Signal Ratio(ISR) of -10dB corresponding to low power and an overall mAP50 performance of 93%.

INDEX TERMS LTE, 5G, anomaly detection, object detection.

I. INTRODUCTION

As the use of wireless devices continues to increase, the risks of signal degradation, interference, and false alarms also rise, making reliable wireless communication a critical concern. In particular, industries that use wireless signals—such as communications, broadcasting, radar, and military applications—implement spectrum monitoring to detect the presence of unwanted signals. LTE and 5G mobile communication bands are characterized by numerous users and high demands for communication stability. Therefore, preventive measures against service quality degradation caused by interference or external anomalous signals are necessary. Moreover, these bands exhibit a high spectrum density and a complex electromagnetic environment compared to other bands, making anomalous signal detection more challenging.

Additionally, there is a limitation in that it can only be applied to specific signal types. Traditional mechanical

algorithms are scenario- and parameter-dependent, meaning that a single algorithm may not be effective in all situations. In contrast, deep learning has the advantage of detecting a wide range of signals, as it can learn from trained signals and adapt to various signal types. Studies in [1] and [2] detect anomalous signals by using manually designed patterns or by converting the problem of spectrum anomaly detection into one of statistical significance testing. Others exploit spatial distributions of received signal strength across different locations to detect anomalies.

Deep learning, on the other hand, can deliver robust performance across various scenarios by leveraging large volumes of data for training. Consequently, it has been employed extensively in spectrum monitoring studies using multiple approaches. For instance, [3], [4], [5], [6] investigated a classification-based network for identifying various modulation types, whereas [7], [8], [9], [10] proposed an unsupervised anomaly detection approach based on an autoencoder. Although both methods achieved high performance, each exhibits clear limitations. Classification-based

The associate editor coordinating the review of this manuscript and approving it for publication was Miguel López-Benítez¹.

networks can only be applied to single modulation signals in a spectrum, making it difficult to identify the range of anomalous signals that can appear in wideband environments like LTE and 5G. Meanwhile, unsupervised autoencoder-based approaches do not rely on labeled data, making them less effective when anomalous signals closely resemble normal signals. Additionally, neither of these methods provides time or frequency information about the detected anomalies, which constrains the comprehensive utilization of signal information.

In contrast, anomaly detection networks based on object detection can localize anomalous signals in both the time and frequency domains, providing multiple layers of information for a single data sample. For this reason, studies in various signal and spectrum scenarios have adopted object detection methods. For example, [11] investigated the use of object detection networks to identify various modulation signals, and [12] used object detection to detect frequency-hopping (FH) signals. However, these studies considered relatively narrowband spectra with limited potential for overlap among multiple signal objects, resulting in relatively simple conditions. Although [13] proposed a network to detect LTE and 5G signals, it focused only on these two communication signals and did not consider additional anomalous signals. Another study [14] addressed the detection of LTE, 5G, and radar signals in mixed spectrograms but employed a standard YOLO v5 network, which may not fully account for the unique properties of signals in the spectrogram. The differences between conventional object detection and RF signal-based object detection can be observed in Figure 1.

Furthermore, [15] used a novel deep learning network that explicitly integrates time-frequency attention mechanisms into spectrum anomaly detection. By collectively modeling temporal and spectral dependencies, the method shows robustness in detecting nonstationary or complex anomalies. However, the performance may degrade when anomalies are subtle within dominant standard patterns due to reconstruction-based detection limits.

This paper proposes a network capable of detecting multiple anomalous signals in complex electromagnetic environments using time-frequency analysis methods. To leverage object detection methods, 1D signal data was transformed into spectrograms with time-frequency axes using the Short-Time Fourier Transform (STFT).

The data used in this study differs from conventional object detection tasks. Specifically, it is based on wireless signals represented as 2D data, where the axes correspond to time and frequency. Unlike typical objects that occlude each other when overlapped, in the case of signals, stronger signals become more prominent, while weaker signals appear faintly. These characteristics are particularly observable in high-spectrum-density environments, such as LTE and 5G. In such cases, anomalous signals overlapping with existing signals are more easily detected. To reflect these unique characteristics, the study explored object detection-based networks that can effectively incorporate the distinctive

features of anomalous signals in spectrograms. Furthermore, a deep learning module was developed to enhance detection performance for low-power signals. The proposed network was compared with multiple object detection networks to verify its suitability for the domain.

Also, this paper focuses on accuracy, while model lightweighting and real-time performance are crucial aspects in spectrum monitoring. However, these will be addressed in future research.

Overall, the novel contributions this paper offers can be summarized as follows:

- 1) Utilization of object detection methods to analyze time and frequency information of signals.
- 2) Proposal of a robust detection framework that maintains high performance under low-ISR (low-power) conditions, which are common in real-world wireless environments.
- 3) Development of a network that effectively reflects the unique characteristics of signal data.

The remainder of this paper is organized as follows. In Section II, we introduce the anomalous spectrum environment and types of anomalous signals used in this study. Section III discusses data preprocessing, which converts IQ data into spectrograms, and describes the proposed deep learning network structure and its object detection approach. In Section IV, we present various experimental parameters and detection performance results, including comparisons with existing object detection networks. Finally, Section 5 offers our conclusions and suggestions for future research directions.

II. METHOD

In this chapter, we describe the Short-Time Fourier Transform (STFT), which converts one-dimensional data into two-dimensional time-frequency image representations, and the architecture of the deep learning network designed to detect anomalous signals within the spectrogram. The dataset used in this study is based on the data from [16]. The dataset was collected from the LTE and 5G frequency bands, containing 128,271 anomalous signals across 85,500 STFT samples. The anomalous signals were generated through Python-based simulations. The spectrogram was generated by first creating a signal of length 1,300,000 with a 3200, 1600 STFT, followed by downsampling to 400, 400. The vertical axis represents time, with each data point corresponding to 0.21 ms, and the frequency bandwidth is 48 MHz. This information is also specified in the Table 2. The Blackman-Harris window was used for the transformation. Even for the same signal, different resolutions in the STFT lead to varying results. Lower resolution increases the signal processing and network processing speed but reduces the resolution of the signal, which in turn degrades anomaly detection performance. On the other hand, higher resolution slows down processing speed but enhances the resolution, potentially improving detection performance. We conducted experiments by selecting parameters that best reflect the



FIGURE 1. The difference between general object detection and spectral object detection. The left image corresponds to a standard photograph, where overlapping cars in the field of view are obscured. The right image corresponds to signals in the spectral domain, where visibility is determined by the power magnitude of the signals when overlapping occurs, and it typically exhibits a longer aspect ratio compared to general objects.

TABLE 1. Network architecture description and parameter details.

Layer name		Layer details	Output Size (H x W x C)
Conv		Convolution: 3x3, 32 channels	400 x 400 x 32
ResNet-18	ResNet_Block1	Deformable conv: 3x3, 32 channels, Convolution: 7x7, 64 channels	200 x 200 x 64
	ResNet_Block2	Deformable conv: 3x3, 64 channels, Convolution: [3x3, 64] x 2	100 x 100 x 64
	ResNet_Block3	Deformable conv: 3x3, 64 channels, Convolution: [3x3, 128] x 2	50 x 50 x 128
	ResNet_Block4	Deformable conv: 3x3, 64 channels, Convolution: [3x3, 128] x 2	25 x 25 x 256
	ResNet_Block5	Deformable conv: 3x3, 64 channels, Convolution: [3x3, 256] x 2	12 x 12 x 512
Upsampling	Up_Block1	Deconvolution: 3x3, 128 Convolution: [3x3, 128] x 2	25 x 25 x 256
	Concat1	Concat(Up_Block1, ResNet_Block5)	50 x 50 x 128
	Up_Block2	Deconvolution: 3x3, 128 Convolution: [3x3, 64] x 2	100 x 100 x 64
	Concat2	Concat(Up_Block2, ResNet_Block4)	200 x 200 x 32
	Up_Block3	Deconvolution: 3x3, 128 Convolution: [3x3, 128] x 2	200 x 200 x 32
	Concat3	Concat(Up_Block3, ResNet_Block3)	100 x 100 x 64
	Up_Block4	Deconvolution: 3x3, 128 Convolution: [3x3, 128] x 2	200 x 200 x 32
	Concat4	Concat(Up_Block4, ResNet_Block2)	200 x 200 x 32
Conv		Convolution: 3 x 3, 64 channels Convolution: 3 x 3, 128 channels	200 x 200 x 64 200 x 200 x 128
Heat map		Convolution: 3 x 3, 3 channels	200 x 200 x 3
HW		Convolution: 3 x 3, 2 channels	200 x 200 x 2
Offset		Convolution: 3 x 3, 2 channels	200 x 200 x 2

signal characteristics of the network band, as described above.

A. SCHEME OVERVIEW

Initially, one-dimensional IQ data is transformed into a two-dimensional spectrogram using the Short-Time Fourier Transform (STFT). Subsequently, to detect anomalous signals, the spectrogram is input into the proposed CenterNet model, which identifies the central points of the anomalous signals and determines the horizontal and vertical dimensions of the bounding boxes. The overall process—including

signal acquisition, signal processing, and the deep learning procedure—can be observed in Figures 3 and 4.

B. STFT

The Short-Time Fourier Transform (STFT) is a method that does not perform a Fast Fourier Transform (FFT) on the entire one-dimensional (1D) IQ signal at once. Instead, it defines a window corresponding to the desired number of frequency bins and continuously applies the FFT by overlapping the current windowed data with the subsequent data by the specified overlap amount. Ultimately, the FFT results are

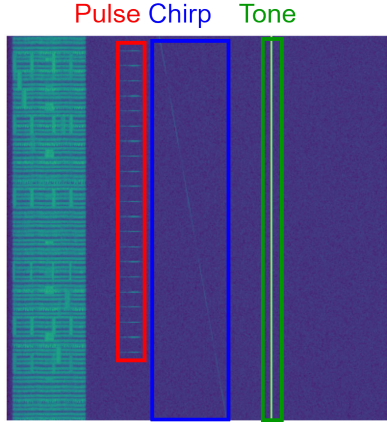


FIGURE 2. Examples of types of generated signals.

TABLE 2. Table summarizing the dataset used, including the number of samples for training and testing, the number of classes, and the spectrogram dimensions.

Train	Test	Abnormal Signal Classes
85,500	104,500	Tone, Chirp, Pulse
Time Duration	Bandwidth	Spectrogram Size
0.21ms	48MHz	400,400

stacked along the time axis. This process is represented by equation (1). $\omega(t)$ denotes the window function, with commonly used window functions including Hann and Gaussian. $x(t)$ represents the signal to be transformed, which, in this study, corresponds to the IQ signal.

$$\text{STFT}\{x(t)\}(\tau, \omega) = \int_{-\infty}^{\infty} x(t) w(t - \tau) e^{-i\omega t} dt \quad (1)$$

After performing the Short-Time Fourier Transform (STFT), the time-frequency characteristics of Tone, Chirp, and Pulse signals can be analyzed. This analysis facilitates the differentiation between these signals and typical LTE and 5G signals, enabling more effective signal.

C. BASE NETWORK

The architecture of the proposed network is depicted in Figure 3, illustrating the integration of the foundational CenterNet network with additional structures designed to facilitate training. The CenterNet network, employed as the base network in this study, consists of three primary components essential for object detection and bounding box generation.

Firstly, backbone networks such as ResNet [17] DLA [18], and VGG16 [19] are utilized for feature extraction. Secondly, a structure for upsampling the extracted features is incorporated, which is necessary to capture both low-dimensional and high-dimensional characteristics present in the spectrogram. These two components work together to ensure comprehensive feature extraction by reflecting the diverse feature scales that may exist within the spectrogram.

Thirdly, the network includes three parallel pathways dedicated to identifying the center points of the signals, determining the width and height of the bounding boxes, and calculating the offsets resulting from downsampling. These pathways enable precise localization and sizing of anomalous signals within the spectrogram.

In this paper, the three modules are referred to as the center point module, the bounding box module, and the offset module, respectively. By centralizing these modules within the network architecture, the proposed model effectively enhances the detection and localization of anomalous signals in complex spectrogram data.

All three modules are represented as heatmaps, and training is conducted by regressing them to the ground truth. The center point module uses a heatmap generated by a Gaussian kernel that reflects the size of the object as the ground truth, with values ranging from 0 to 1. This is mathematically expressed in equations (2) through (4). equation (2) represents the input data, where I denotes the input image, $R^{(W \times H \times 1)}$ indicates the size of the input image, W, H correspond to the width and height of the image respectively, and the subsequent number represents the number of channels in the input image. In this paper, a one-channel spectrogram corresponding to size was used, so the number is 1.

$$I \in \mathbb{R}^{W \times H \times 1} \quad (2)$$

Equation (3) represents a Gaussian kernel that defines the center of an object as the ground truth. This equation generates a Gaussian-shaped heatmap based on $(\hat{p}_x, \hat{p}_y)$, which denotes the coordinates of the object's center. The values of the heatmap vary according to the distance from the object center, as illustrated by $(x - \hat{p}_x)^2 + (y - \hat{p}_y)^2$. The variance of the Gaussian, σ_p^2 , is adjusted according to the size of the object, with larger objects resulting in a broader Gaussian distribution. Specifically, at a distance far from the center point, $Y_{x,y,z} = 0$, it indicates the absence of an object, whereas if the coordinates coincide with the center point, $Y_{x,y,z} = 1$, it signifies the presence of an object at that location.

$$Y_{x,y,z} = \exp\left(-\frac{(x - \tilde{p}_x)^2 + (y - \tilde{p}_y)^2}{2\sigma_p^2}\right) \quad (3)$$

Equation (3) represents the output of the center point module, where Y denotes the output data and $[0, 1]^{\frac{W}{S} \times \frac{H}{S} \times C}$ indicates the shape of the output data, which ranges from 0 to 1. S signifies the downsampling factor, and in this paper, a value of 2 is used, resulting in a final output size of 200×200 . Since the center point module must perform both localization and classification functions simultaneously, channel C is present. The dataset proposed in this paper consists of three classes—tone, chirp, and pulse—and is consequently configured with three channels.

$$Y \in [0, 1]^{\frac{W}{S} \times \frac{H}{S} \times C} \quad (4)$$

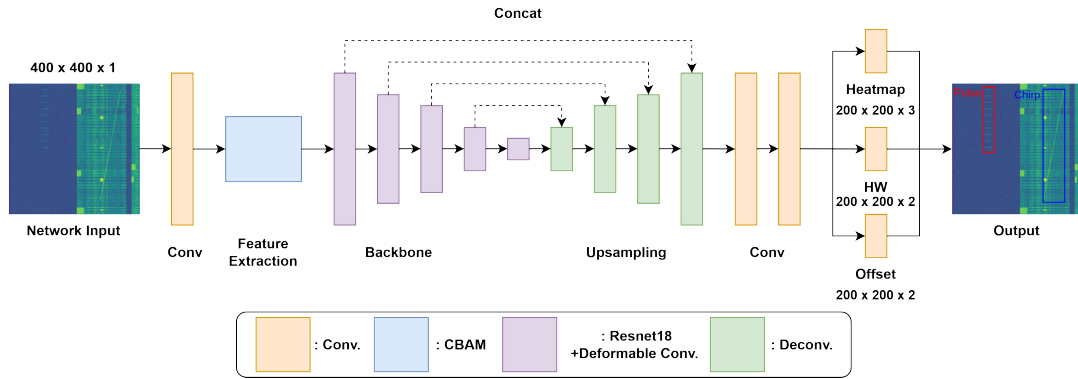


FIGURE 3. The structure of the proposed network. It is based on the CenterNet architecture, with an additional structure combining convolution, concatenation, and Deformable Convolution.

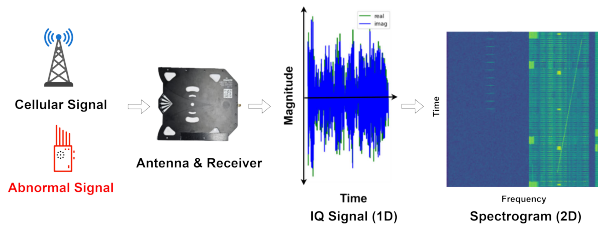


FIGURE 4. The signal processing flow begins with the reception of external wireless signals through a receiver. The acquired IQ signals are then transformed into spectrograms using the STFT. These spectrograms serve as input to a deep learning network, which performs anomaly detection.

Equation (5) represents the focal loss function for the center point module. This loss function is designed to address imbalanced data and focus on challenging samples. In the loss function, α is a parameter that reduces the loss for easy samples while emphasizing the loss for difficult samples, and β is a weight that adjusts the influence of the background pixel loss. α is set to 1, and β is set to 3. Y_{xyc} represents the ground truth heatmap, indicating whether a pixel corresponds to class c . $\hat{Y}_{xyc}$ is the heatmap predicted by the model, representing the probability of class c at each x, y coordinate. N is the normalization coefficient, corresponding to the number of actual objects. It serves to normalize the loss by dividing it by the number of objects in the heatmap.

$$L_{pt} = -\frac{1}{N} \sum_{x,y,c} \ell_{xyc}, \quad (5)$$

$$\ell_{xyc} = \begin{cases} (1 - \hat{Y}_{xyc})^\alpha \log(\hat{Y}_{xyc}), & \text{if } Y_{xyc} = 1, \\ (1 - Y_{xyc})^\beta (\hat{Y}_{xyc})^\gamma \log(1 - \hat{Y}_{xyc}), & \text{otherwise.} \end{cases} \quad (6)$$

The offset module serves to correct the positions of objects predicted by the center point module. Since the final output size is reduced by a factor of S , offsets due to downsampling may occur. This module is employed to compensate for such offsets by determining the offset values using an L1 loss function. The corresponding equation is presented in

equation (6). Here, $\tilde{O}_p$ represents the offset values predicted by the model, and M denotes the number of actual object centers present in the heatmap, serving to normalize the loss values. The remaining variables are defined as in the center point module. The values of the heatmap range from 0 to 1, which indicate the magnitude of the offset relative to the size of the output data.

$$L_{off} = \frac{1}{M} \sum_p \left| \hat{O}_p - \frac{p}{S} - \tilde{p} \right| \quad (7)$$

The bounding box module is responsible for determining the size of the bounding box corresponding to an object. In equation (8), p_m represents the center coordinates of the bounding box, $(x_1^{(m)}, y_1^{(m)})$ denotes the top-left coordinates of the bounding box, and $(x_2^{(m)}, y_2^{(m)})$ indicates the bottom-right coordinates.

$$p_m = \left(\frac{x_1^{(m)} + x_2^{(m)}}{2}, \frac{y_1^{(m)} + y_2^{(m)}}{2} \right) \quad (8)$$

Therefore, the size s_m of the bounding box is defined as shown in equation (9).

$$s_m = (x_2^m - x_1^m, y_2^m - y_1^m) \quad (9)$$

Regression on the heatmap is performed using the offset function with the aforementioned parameters. Similarly, the loss function employed is the L1 loss function, as defined in equation (10).

$$L_{size} = \frac{1}{M} \sum_{m=1}^M \left| \hat{S}_{p_m} - S_m \right| \quad (10)$$

The final loss function employed in the model consists of the sum of three individual loss functions. In the model utilized in this paper, the loss function coefficient λ_{size} corresponding to the bounding box module was set to 0.1. The equation is presented in equation (11).

$$L_{train} = L_{pt} + \lambda_{size} L_{size} + L_{off} \quad (11)$$

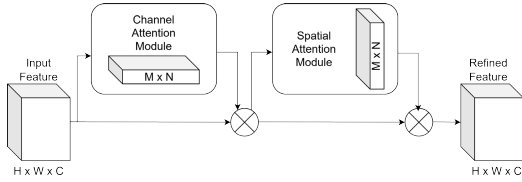


FIGURE 5. The CBAM network architecture utilizes both channel and spatial attention, enabling the model to assign weights to informative features during training.

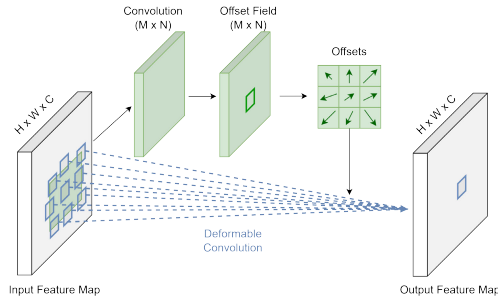


FIGURE 6. The Deformable Convolution network architecture learns offset information instead of relying on a fixed convolutional kernel, allowing for adaptive spatial sampling.

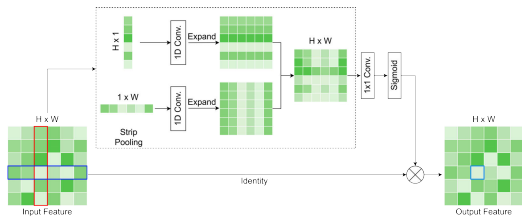


FIGURE 7. The Strip Pooling network architecture employs 1D convolution along the horizontal and vertical axes of the feature map, allowing the model to capture global contextual information in both directions.

D. SIGNAL FEATURE EXTRACTION

The Convolutional Block Attention Module (CBAM) [20] is a method that applies attention mechanisms to both the channel and spatial dimensions of feature maps. Unlike the two methods introduced previously, CBAM extracts signal features through attention mechanisms, emphasizing important features present in the feature map while suppressing unnecessary ones. These characteristics enable the effective learning of channels and spatial locations where attention should be focused within the image. The architecture of this network can be found in Figure 5.

Deformable Convolution [21], in contrast to conventional convolution, can dynamically adjust the sampling positions of the kernel during feature learning by moving the kernel. This capability allows the model to learn information from the surrounding offsets, making it robust to various shapes, rotations, and scale changes, thereby enabling more flexible learning of feature maps. For these reasons, Deformable Convolution was selected as it is considered suitable for learning to detect signals with horizontal, vertical, and

diagonal structures. Signals exist in the spectrogram with various durations and frequencies. Additionally, depending on the signal's detailed parameters, the slope and the area it occupies in the time-frequency space can vary. In particular, for signals with strong diagonal characteristics, such as chirp, the actual signal region is much smaller compared to the area defined by the bounding box. This implies that using fixed, rectangular-shaped convolutions limits the amount of information that can be learned. In contrast, Deformable Convolution, which differs from regular convolution by having a variable kernel shape, is more effective because it allows for the learning of more information even in the same context. The structure of this network is described in Figure 6.

Signals present in a spectrogram predominantly exhibit linear structures. This characteristic applies not only to LTE and 5G signals but also to the proposed dataset types, including tone, chirp, and pulse signals. Strip Pooling [22] is a network designed to effectively capture information along both the horizontal and vertical axes. As illustrated in Figure 7, it performs 1D convolution along the respective axes of the feature map to extract relevant information, which is then expanded back into 2D and fused for further processing.

The architecture of the final network used is illustrated in Figure 3. Through CBAM, the feature map passing through the first convolution is weighted in both the channel and spatial dimensions using the attention mechanism to highlight characteristic regions. ResNet18 was employed as the backbone network, and when upscaling the feature maps, feature maps of the same size from ResNet were concatenated to prevent the loss of information from the previous feature maps. Additionally, Deformable Convolution was applied at each layer of ResNet to learn information from offsets beyond the kernel size. After passing through additional convolutions, a detection module was incorporated to predict the bounding boxes of objects. The parameters used in the network implementation are presented in Table 1.

III. RESULTS AND DISCUSSION

In the previous chapter, several modules facilitating the extraction of signal features were introduced. This chapter details the parameters used for training and preprocessing and conducts a performance comparison of the modules referenced in CenterNet. Additionally, performance comparisons with various versions of YOLO, the transformer-based detection model RT-DETR, and the two-stage detector Libra R-CNN are also presented.

A. EXPERIMENT SETTINGS

The parameters used for training were a batch size of 64, 100 epochs, and a confidence threshold of 0.25. The Adam optimizer was employed with a learning rate of 0.001 and a weight decay of 0.001. Additionally, a CosineAnnealing-WarmRestarts scheduler, which combines Cosine Annealing and Warm-Up techniques, was utilized with the following parameters: $T_0 = 25$, $T_{mult} = 1$, $\eta_{max} = 0.1$, $T_{up} = 10$, and $\gamma = 0.5$.

In this paper, Average Precision (AP) was used to evaluate object detection performance. AP comprises Precision, which is the ratio of correctly predicted positive instances to the total predicted positives, and Recall, which is the ratio of correctly predicted positive instances to all actual positives. The computation involves plotting the Precision-Recall curve, which varies with the confidence threshold, and calculating the area under the curve. A larger area indicates better detection performance. When evaluating multiple types of objects simultaneously, the mean Average Precision (mAP) is employed, which is calculated by averaging the APs of all object types.

B. EFFECTS OF FEATURE EXTRACTION MODULES

The feature modules described in Section III-B each extract features in different ways. Therefore, an ablation study was conducted to determine which module is most suitable for signal feature extraction. The modules and their combinations used in the ablation study are presented in Table 3.

Figure 8 and Table 4 shows the training results based on a single sample, and among the various combinations of modules, the single use of Deformable Convolution exhibited the highest performance. Notably, while the model using only Deformable Convolution attained APs of 93.9% and 93.7% for Tone and Chirp respectively—slightly lower than the 94.2% and 94% of Type 6 (i.e., Deformable Convolution plus Strip Pooling)—it outperformed Type 4 (93%) by scoring 91.3% AP in Pulse. These findings indicate the contributions of each feature extraction module to signal detection.

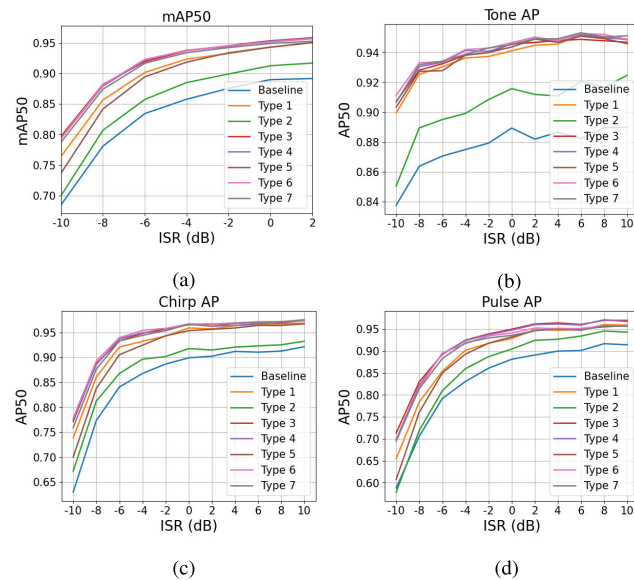


FIGURE 8. AP performance for each feature extraction module combination. In order of (a), (b), (c), and (d), the graph represents mAP performance as well as the AP performance for the tone, chirp, and pulse classes.

Although all single-module cases did lead to performance gains, Strip Pooling yielded the lowest performance, and CBAM achieved the second-highest. These results suggest

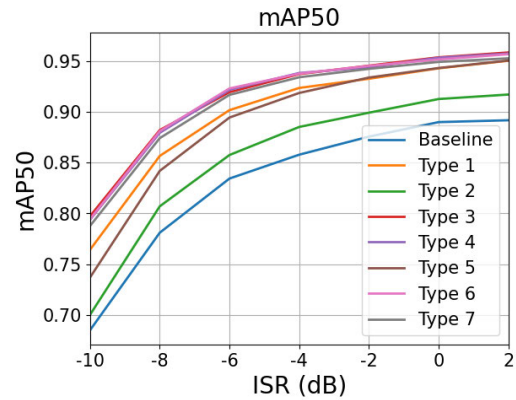


FIGURE 9. The results illustrating the performance of various combinations of feature extraction modules in the range of -10 to 0 dB demonstrate that cases incorporating deformable convolution achieve superior performance.

that, although Strip Pooling is advantageous for learning global horizontal and vertical information, it may not adequately capture local details. The application of CBAM's spatial and channel attention strengthens the adaptive feature refinement capabilities. Signals in the time-frequency space generally exhibit linear and rectangular characteristics, resulting in relatively simple shapes. However, unlike conventional images, the wireless spectrum shows high variability in signal characteristics due to noise and signal overlap. Even similar signals can display different properties depending on the background band. The actual training and evaluation results confirmed that CBAM's adaptive feature refinement properties were effectively utilized to emphasize and suppress domain-specific features of the wireless spectrum. As a result, it achieved the second-highest performance among the three networks. On the other hand, Deformable Convolution recorded the highest performance overall, suggesting that it best reflected both the linear and power characteristics of the signal. Additionally, certain module combinations outperformed Deformable Convolution alone for specific ISRs or signal types. Notably, Type 6 recorded a mAP only 0.1% lower than Type 2, while outperforming it by 0.3% in both Tone and Chirp, but falling behind by approximately 1% in Pulse. When all three modules were used, the performance was lower than that of using one or two modules, confirming that more modules do not necessarily lead to better performance. Through this ablation study of module combinations, it was demonstrated that performance in terms of mAP and AP for each class varies depending on the features each module captures.

From Figure 9, it can be observed that the scale of the graph is relatively small, reflecting only lower ISR values, yet multiple networks demonstrate similar performance. Notably, networks incorporating Deformable Convolution exhibit this trend. Despite combining two feature extraction modules, Type 5, which does not include Deformable Convolution, shows relatively lower performance. This is particularly

TABLE 3. Module configurations and computational cost comparison across architectures. The usage of CBAM, deformable convolution, and strip pooling modules is listed along with total parameters, FLOPs, and parameter size.

Architecture	CBAM	Deformable Convolution	Strip Pooling	Total Parameters (M)	FLOPs (G)	Parameter Size (MB)
Baseline [23]	X	X	X	14.12	24.98	56.10
Type 1 [20]	O	X	X	38.50	68.38	153.98
Type 2 [21]	X	O	X	17.16	20.23	55.41
Type 3 [22]	X	X	O	14.16	26.26	56.10
Type 4	O	O	X	17.16	20.27	55.42
Type 5	O	X	O	13.73	18.32	54.90
Type 6	X	O	O	17.16	20.43	55.43
Type 7	O	O	O	17.17	20.47	55.43

TABLE 4. Performance comparison table based on network module combinations. Overall performance and performance for each class are recorded. The results are based on a single test subset and do not reflect statistical variance.

Architecture	mAP ₅₀ (%)	AP ₅₀ ^{tone} (%)	AP ₅₀ ^{chirp} (%)	AP ₅₀ ^{pulse} (%)
Baseline [23]	85.7	87.6	85.9	83.4
Type 1 [20]	92.1	93.7	92.8	89.7
Type 2 [21]	93.1	93.9	93.7	91.6
Type 3 [22]	88	90.3	88	85.7
Type 4	93	94	93.7	91.3
Type 5	91.3	93.8	91.6	90.3
Type 6	93	94.2	94	90.6
Type 7	92.6	94.1	93.5	90.3

evident as its performance falls below that of using CBAM alone.

Tone signals generally exhibit high detection performance (approximately 90%) even under low-power conditions, with the smallest performance increase as power levels rise. Although this suggests that tone signals might be the easiest to detect, their performance under high-power conditions is approximately 1% lower than that of chirp and pulse signals, indicating a similar level of overall difficulty. Furthermore, tone signals show the smallest performance variance across different module configurations, which can be attributed to their inherently simple signal structure.

In contrast, chirp signals display a more noticeable gap in performance between low- and high-power conditions. Although their detection accuracy improves as the power level increases, the rate of improvement steadily declines starting from -4 dB, implying minimal difference in performance between moderately powered and high-powered chirp signals. A similar trend emerges when comparing different module configurations, where discrepancies in detection performance are more pronounced at lower power levels. Nevertheless, when modules are employed, chirp signals can achieve detection rates exceeding 95% in high-power scenarios, leading to an overall level of difficulty comparable to tone signals despite lower performance at low power.

Pulse signals exhibit an even greater disparity in detection performance between low- and high-power conditions than chirp signals. Their performance improves more gradually and appears to saturate around 2 dB. Although performance can reach approximately 95% under high-power conditions when modules are utilized, the high difficulty of detecting

pulse signals at low power renders them the most challenging among the three classes.

To validate the reliability of performance differences between model variants, we performed pairwise statistical significance tests using the paired *t*-test on the *mAP* values from 11 randomized test splits per model. The distribution of *mAP* scores across different ablation settings is illustrated in Figure 10 (a).

Moreover, we also evaluated the computational efficiency of each network architecture by measuring the average inference time per batch. Among all networks, Type 1 had the fastest inference time of 3.5 ms per batch with the fewest parameters (13.73M), indicating its efficiency advantage. However, its *mAP* was lower at 85.7%.

In contrast, Type 2, which uses only Deformable Convolution, achieved the best balance with the highest *mAP* of 93.1% and a reasonable inference time of 4.2 ms per batch and parameter size (17.16M). Meanwhile, Type 7 showed the slowest inference time (5.3 ms per batch) and did not outperform Type 2, indicating that adding more modules does not necessarily improve performance or efficiency. These results, visualized in Figure 10 (b), support that Type 2 is the most well-balanced model for real-time anomaly detection in wireless communication systems. Additionally, Figure 10 (c) compares model-wise *mAP* against inference time, highlighting that Type 2 achieves the best trade-off, while Type 5 provides the most efficient configuration with the lowest latency and competitive *mAP*.

TABLE 5. Updated average *mAP*, inference time, and their ratio across model types. Values are based on 11 randomized test subsets.

Architecture	Avg. mAP (%)	Avg. Inference Time (ms)	mAP / Time
Baseline	85.71	3.162	27.11
Type 1 [20]	91.98	5.393	17.06
Type 2 [21]	93.06	4.342	21.43
Type 3 [22]	88.06	3.682	23.92
Type 4	92.86	4.860	19.11
Type 5	91.23	3.572	25.54
Type 6	92.95	4.820	19.28
Type 7	92.61	5.100	18.16

To provide a more comprehensive assessment as suggested, we also analyzed the efficiency of improvement per cost increase, defined as the ratio of performance gain (*mAP*) to the increase in inference time, i.e., *mAP* / Inference

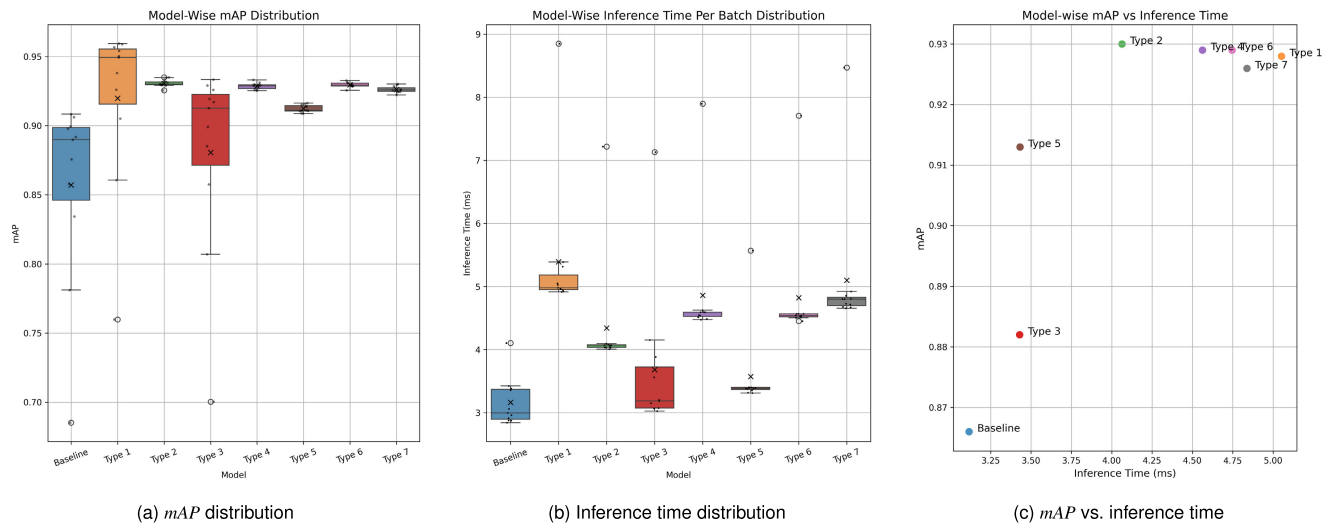


FIGURE 10. Statistical comparison across 11 randomized test subsets, visualizing model performance variation and trade-offs.

Time. This metric assesses how effectively a model improves accuracy while minimizing the sacrifice of speed. As a result, Type 5 showed the highest score in this ratio, providing the most efficient improvements in terms of performance and cost.

Moreover, we evaluated the overall balance between accuracy and speed by analyzing the average mAP and inference time directly. According to this evaluation, Type 2 achieves the highest average mAP among all models while maintaining reasonable inference time, confirming that it is the most balanced model in terms of overall detection performance.

These results suggest that when real-time responsiveness is the top priority, Type 5 is the most favorable architecture due to its efficient trade-off between performance and computational cost. On the other hand, Type 2 is the best architecture when accuracy is the primary concern. A detailed comparison of computational costs for all model types is presented in Table 3.

C. COMPARISON WITH GENERALIST MODELS

To evaluate the performance of the network developed in this study, we compared it against YOLOv6, YOLOv8, and YOLOv11 [24], [25]. All models were selected as small models, as their parameter sizes are similar to those presented in the paper, ranging from approximately 10M to 20M. Analyzing the AP graphs shown in Figure 11 and Table 6 reveal that, overall, Tone is the easiest class and Pulse is the most challenging. Among the YOLO variants, YOLOv8 demonstrated the highest performance in all classes. Meanwhile, DETR achieved 96.1% for Tone—its best result—but showed a lower performance of 91.1% for Chirp compared to YOLOv8 and YOLOv11. In contrast, the proposed network attained the highest performances of 93.7% for Chirp and 91.3% for Pulse, and from the analysis

of the graphs, it appears to excel particularly on low-power signals. When averaging across the three classes to obtain an mAP, the proposed network also achieved the highest score of 93.1%.

For the tone class, as observed in the module performance comparison, the increase in detection performance with higher power levels remains relatively small. However, when compared with the other classes, a more pronounced performance gap among different networks becomes evident. In addition, both chirp and pulse signals exhibit a saturation in performance gains at certain power levels, consistent with the observations from the module performance comparison. Notably, the base network equipped with Deformable Convolution demonstrates a substantially larger performance gap relative to the other networks.

In summary, while its performance on Tone was approximately 2.1% lower than RT-DETR, it surpassed RT-DETR in the other classes. Its strong detection capability even at low power suggests that the network effectively captures the linear features of signals. Specifically, it likely recognizes signals even when they are masked or truncated under low-power conditions. Consequently, the feature-extraction modules considered in this study remained effective even when compared against other well-known models, and the proposed network showed the best performance overall, rendering it highly suitable for spectrum monitoring.

Detection performance is most pronounced at lower power levels, indicating that detection performance in low-power conditions is crucial for evaluating detection performance. However, all three signals have similar detection performance when the power is high. Therefore, we need to focus on low-power scenarios for detailed analysis. Tone signals, which are the simplest type, exhibit minimal shape change even as parameters such as bandwidth and periodicity vary. As a result, despite operating at low power, tone signals likely

achieve higher detection performance because they are highly similar to each other. Chirp signals, however, display diagonal patterns in spectrograms that change significantly with parameter variations, leading to lower detection performance than tone signals under low-power conditions.

To better understand how object detection models handle this different detection challenge, we included Libra R-CNN for comparison. Libra R-CNN features a balanced feature pyramid (BFP) and a combined sampler that addresses the limitations of typical FPNs. The BFP effectively fuses multi-scale information by using both low- and high-dimensional features, which helps detect small or faint signals with low-ISR. We selected this model because of its expected advantage in detecting low-ISR or small signals. We also designed to handle spectrogram-based abnormalities—such as tone, chirp, and pulse—where class distribution is imbalanced, thanks to its combined sampler that considers class imbalance and sampling difficulty. Models like CenterNet and YOLO employ an anchor-free approach that detects abnormal signals by directly predicting the object center through a center point-based heatmap, without the need for predefined anchors. In contrast, Libra R-CNN uses a two-stage, anchor-based Region Proposal Network (RPN). We chose it to compare how Libra R-CNN performs relative to anchor-free detectors and to see how the limitations of anchor-based methods affect spectrogram-based anomaly detection.

Non-local operations in the BFP enhance broad contextual information in the time-frequency domain, allowing for the effective detection of signals with wide and regular patterns, such as chirps. Pulse signals, which show as sharp energy bursts over short periods, are also captured more reliably because BFP integrates features across various resolutions. Despite these strengths, Libra R-CNN achieved a lower mAP than CenterNet and YOLO models. This is likely because, as an anchor-based RPN, it struggles to accurately match small, narrow signals, such as tones, due to predefined anchor sizes. Fixed anchor sizes limit its ability to detect diverse abnormal signals in the spectrum, particularly those that are small or dense signals. For instance, narrow and weak patterns, such as tone signals, do not align with the predefined anchors, leading to limited detection performance.

Although Libra R-CNN attained APs of 66.2% and 57.3% for Chirp and Pulse respectively, but only 20.7% for Tone - indicating a significant drop in performance for smaller signals. Overall, the fixed anchor sizes of the anchor-based approach limit its ability to detect all types of abnormal signals in the spectrum, particularly reducing recall for small or densely crowded signals.

D. SIGNAL CLASS PERSPECTIVE ANALYSIS

An ablation study and comparisons of various deep learning models revealed that, despite the relatively simple shapes of all three signal types, observable differences exist in their detection performance. These differences become particularly pronounced at lower power levels, indicating that

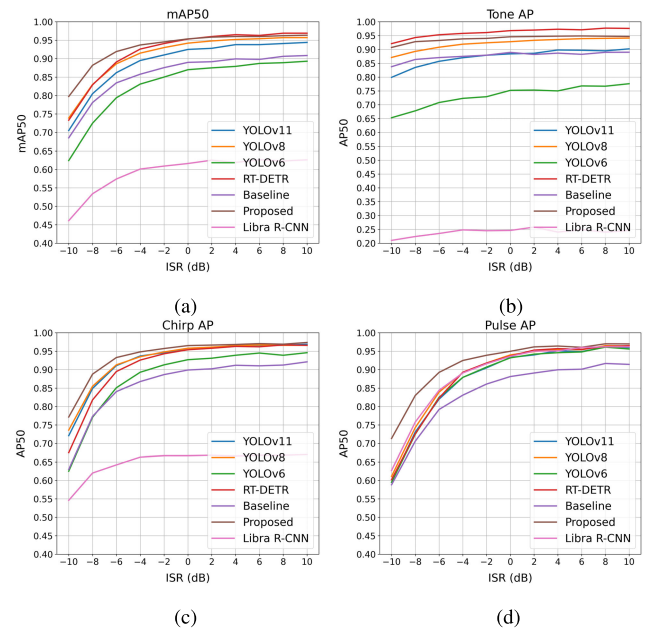


FIGURE 11. Performance comparison with other networks. In order of (a), (b), (c), and (d), the graph represents mAP performance as well as the AP performance for the tone, chirp, and pulse classes.

TABLE 6. Table comparing the performance of different network architectures. Comparisons were made with the baseline CenterNet, RT-DETR, Libra R-CNN and YOLO.

Architecture	mAP ₅₀ (%)	AP ₅₀ ^{tone} (%)	AP ₅₀ ^{chirp} (%)	AP ₅₀ ^{pulse} (%)
Proposed	93.0	94.0	93.7	91.3
Libra R-CNN [26]	60.9	23.9	68.8	90.0
RT-DETR [27]	91.8	96.1	91.1	88.1
CenterNet [23]	85.7	87.6	85.9	83.4
YOLOv11-s [25]	89.0	87.3	92.2	87.4
YOLOv8-s [24]	91.0	92.1	92.4	88.4
YOLOv6-s [28]	82.8	73.2	88.0	87.4

detection rates under low-power conditions are crucial in determining the difficulty of signal detection. By contrast, when the power level is high, all three signals exhibit similar detection rates, underscoring the importance of prioritizing low-power scenarios for more detailed analysis.

Tone signals, which represent the simplest form of these signals, exhibit minimal shape changes even when parameters such as bandwidth and periodicity vary. Consequently, despite operating at low power, tone signals likely achieve higher detection performance due to their high similarity to one another.

In the case of chirp signals, the spectrogram reveals diagonal patterns that vary considerably according to parameter changes, resulting in lower detection performance than that of tone signals in low-power conditions. However, chirp signals differ substantially from the real LTE and 5G band signals, and thus are not considered the most challenging class. In fact, under high-power conditions, chirp signals can outperform tone signals.

Pulse signals, on the other hand, manifest horizontally along the frequency axis and similarly exhibit notable shape

variability with parameter changes. They most closely resemble LTE and 5G signals, which feature repeated horizontal patterns in the frequency domain. This resemblance appears to be the main factor contributing to the lowest detection performance among the three signal types in low-power conditions.

IV. CONCLUSION

This work proposes a new deep-learning network for anomaly signal detection in wideband spectrum environments that include LTE and 5G bands. Unlike previous research—often limited to narrower bandwidths or simpler signal compositions—this study designed a dataset reflecting the complex LTE and 5G signal conditions that can arise in real-world settings. Moreover, the dataset follows an object-detection format, providing a foundation for distinguishing temporal and frequency positions of signals as well as their types.

In designing the network, we improved performance by integrating modules effective at feature extraction—Strip Pooling, CBAM, and Deformable Convolution—into the object detection network CenterNet. Notably, the network combined with Deformable Convolution achieved an mAP50 of 93.1%. This surpassed the performance of YOLO and RT-DETR. While it was marginally weaker on the Tone class, it outperformed in other classes, resulting in an overall excellent mAP. Additionally, its strong performance in detecting low-power signals further indicates its suitability for signal detection.

This paper designs a network for detecting anomalous signals within cellular communication bands and achieves improved detection performance for low-power signals. However, there are various other challenges in spectrum monitoring. For real-time spectrum monitoring, fast processing speeds, meaning network lightweighting, are required. Additionally, to achieve general performance across diverse bands, it is essential to develop detection methods that can achieve high performance regardless of the background of the band. Future research will focus on optimizing real-time processing speeds, extending performance validation to various frequency bands, including GNSS and DTV, and advancing signal detection methods that are independent of the band background.

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